东北大学

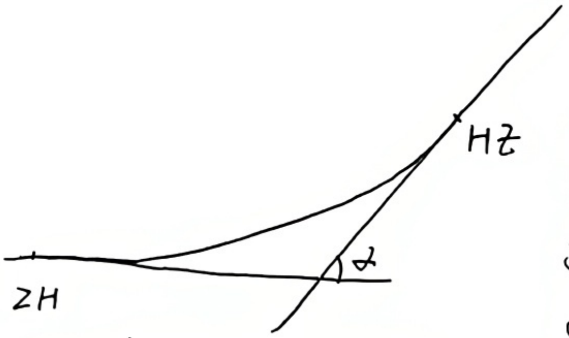
工程测量学课程作业

序次（2）

成绩：　　　　　　　教师：徐忠印　　　　　　批阅日期：

资源与土木工程学院测绘工程专业2023年级2301班

姓名：____　　　　　　　　　　　　　　　学号：________



已知：半径 $R = 4500m$　缓和曲线长 $l_0 = 360m$.

ZH坐标：4554041.588，80485715.572

HZ坐标：4554927.435，80486859.342

JD坐标：4554414.797，80486341.451

$\alpha = 13°54'02''$　JD里程 $DK_{209+271.13}$

(1) 计算缓和曲线参数：

$$\beta_0 = \frac{l_0}{2R} \cdot \frac{180°}{\pi} = 2°17'30.6''$$

$$m = \frac{l_0}{2} - \frac{l_0^3}{240R^2} = 179.990m.$$

$$p = \frac{l_0^2}{24R} = 1.2m.$$

(2) 计算缓和曲线要素：

$$T = m + (R+p) \cdot \tan\frac{\alpha}{2} = 728.703m$$

$$L = 2l_0 + \frac{(\alpha - 2\beta_0) \cdot \pi R}{180°} = 1451.747m$$

$$E = (R+p) \cdot \sec\frac{\alpha}{2} - R = 34.522m$$

$$q = 2T - L = 5.659m.$$

(3) 缓和曲线主点里程计算

$ZH = JD - T = DK_{208+542.427}$ m.

$HZ = ZH + L = DK_{209+994.174}$ m

$YH = HZ - l_0 = DK_{209+634.174}$ m

$HY = ZH + l_0 = DK_{208+902.427}$ m

$QZ = ZH + \frac{L}{2} = DK_{209+268.300}$ m

(检核)

$HZ = JD + T - q = DK_{209+994.174}$（同上，检核成功）

(4) ① ZH < DK208+660.000 < HY 在缓和曲线上.

$l_1 = 660 - 542.427 = 117.573m$

$$\begin{cases} x_1 = l_1 - \dfrac{l_1^5}{40R^2 l_0^2} = 117.573m. \\ y_1 = \dfrac{l_1^3}{6Rl_0} = 0.167m. \end{cases}$$

② HY < DK208+960.000 < QZ 在圆曲线上.

$l_2 = 960 - 542.427 = 417.573m.$

$\theta_2 = \dfrac{l_2 - 0.5l_0}{R} = 3°1'29.54''$

$$\begin{cases} x_2 = m + R\sin\theta_2 = 417.452m \\ y_2 = P + R(1-\cos\theta_2) = 7.470m \end{cases}$$

③ QZ < DK209+360 < YH 在圆曲线上.

$l_3 = 994.174 - 360 = 634.174m$

$\theta_3 = \dfrac{l_3 - 0.5l_0}{R} \cdot \dfrac{180°}{\pi} = 5°46'57.8''$

$$\begin{cases} x_3 = m + R\sin\theta_3 = 633.393m \\ y_3 = P + R(1-\cos\theta_3) = 24.100m \end{cases}$$

④ YH < DK209+760.000 < HZ 在缓和曲线上.

$l_4 = 994.174 - 760 = 234.174m$

$$\begin{cases} x_4 = l_4 - \dfrac{l_4^5}{40R^2 l_0^2} = 234.167m \\ y_4 = \dfrac{l_4^3}{6Rl_0} = 1.321m \end{cases}$$

计算坐标方位角: $\alpha_1 = \alpha_{ZH-JD} = \arctan\dfrac{\Delta Y_{ZH-JD}}{\Delta X_{ZH-JD}} = 59°11'33.2''$

$\alpha_2 = \alpha_{HZ-JD} = \arctan\dfrac{\Delta Y_{HZ-JD}}{\Delta X_{HZ-JD}} = 225°17'31.4''$

计算坐标：

① $\begin{bmatrix} X_1 \\ Y_1 \end{bmatrix} = \begin{bmatrix} X_{ZH} \\ Y_{ZH} \end{bmatrix} + \begin{bmatrix} \cos\alpha_1 & \sin\alpha_1 \\ \sin\alpha_1 & -\cos\alpha_1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 4554101.947 \\ 80485816.470 \end{bmatrix} m$

② $\begin{bmatrix} X_2 \\ Y_2 \end{bmatrix} = \begin{bmatrix} X_{ZH} \\ Y_{ZH} \end{bmatrix} + \begin{bmatrix} \cos\alpha_1 & \sin\alpha_1 \\ \sin\alpha_1 & -\cos\alpha_1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 4554261.804 \\ 80486070.294 \end{bmatrix} m$

③ $\begin{bmatrix} X_3 \\ Y_3 \end{bmatrix} = \begin{bmatrix} X_{HZ} \\ Y_{HZ} \end{bmatrix} + \begin{bmatrix} \cos\alpha_2 & -\sin\alpha_2 \\ \sin\alpha_2 & \cos\alpha_2 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 4554498.975 \\ 80486392.234 \end{bmatrix} m$

④ $\begin{bmatrix} X_4 \\ Y_4 \end{bmatrix} = \begin{bmatrix} X_{HZ} \\ Y_{HZ} \end{bmatrix} + \begin{bmatrix} \cos\alpha_2 & -\sin\alpha_2 \\ \sin\alpha_2 & \cos\alpha_2 \end{bmatrix} \begin{bmatrix} x_4 \\ y_4 \end{bmatrix} = \begin{bmatrix} 4554763.639 \\ 80486691.990 \end{bmatrix} m.$

故有：DK208+660.000的坐标为(4554101.947，80485816.470)

DK208+960.000的坐标为(4554261.804，80486070.294)

DK209+360.000的坐标为(4554498.975，80486392.234)

DK209+760.000的坐标为(4554763.639，80486691.990)